

T41-2 RF POWER AMPLIFIER

INPUT

J2 5-1814832-1

T1 Transformer_1P_5S

1 TURN TWISTED PAIR #26

2 TURNS #26

BN-43-302

SHUTDOWN THERMOSTAT

R10 0

R9 56k

U2 MCP9509HT-E/OT

SET GND

VDD

HYST

*OUT

PWR_FLAG

HYSTERISIS SET 2 DEGREES

10 VOLT BIAS REGULATOR

U3 MCP1804x-A002xOT

VIN

SHDN

VOUT

10V

C20 1uF

C14 0.1uF

+13.4 VOLTS IN

J5

F1 3587TR

E2 28F0121-0SR-10

12V_FIL

E3 28F0121-0SR-10

12V_PA

R19 10

E1 28F0121-0SR-10

C17 0.1uF

PWR_FLAG

D1 Z3SMCxxx

C12 10uF

C3 10uF

C10 47uF

C13 10uF

C2 10uF

C9 47uF

RD16HHF1 TO-220

Q1 Q_NMOS_GSD

Q3 Q_NMOS_GSD

R1 1

R2 1

R5 51

R6 51

R7 51

R8 51

R11 2k

C5 0.1uF

C6 0.1uF

C7 0.1uF

C8 0.1uF

C1 0.1uF

C4 0.1uF

C16 0.1uF

C18 TUNING

T2 Transformer_1P_1S

T3 Transformer_1P_1S

PRIMARY 1 TURN #22

SECONDARY 4 TURNS #22

BN-43-202

OUTPUT

J1 5-1814832-1

Note: C10 and C9 electrolytic capacitors are used for higher voltage supply tests. In this case, delete C2, C3, C12, C13. If using 13.4 volt supply exclusively, do not place C10 and C9.

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Title: T41-2 RF Power Amplifier

Size: USLetter Date: 2025-07-05

KiCad E.D.A. 10.0.4

Rev: 1.7

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